

## **METABOLIC DYSFUNCTION–ASSOCIATED STEATOTIC LIVER DISEASE**

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## INTRODUCTION

Since 2023, metabolic dysfunction-associated steatotic liver disease (MASLD) is the new nomenclature used to describe the condition formerly known as non-alcoholic fatty liver disease (NAFLD) or more commonly “fatty liver” to better reflect our understanding of this liver disease. MASLD lies on the spectrum of liver disease best illustrated by the figure below.

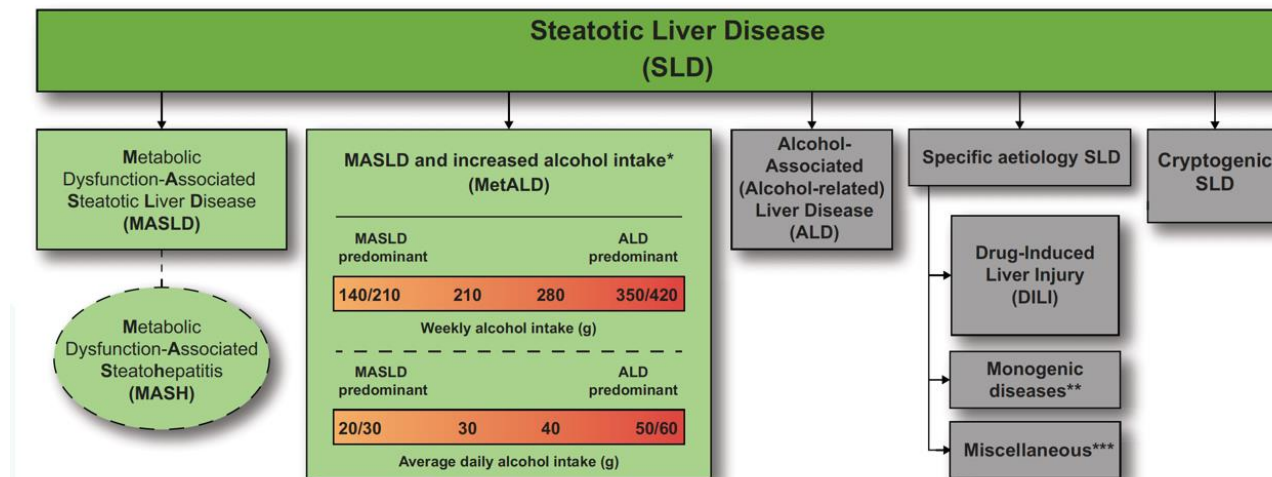


Figure 1: Adapted from America Association for the Study of Liver Disease

## Importance

Most patients with MASLD are asymptomatic and receive their medical care in the primary care sector. Due to the multi-systemic nature of the disease with multiple metabolic associations, patients with earlier stages of the disease are best managed in the primary care setting. It is pertinent that primary care physicians recognize and understand the disease so that early case identification, management and prognostication can be carried out to mitigate the huge morbidity and costs associated with the disease.

## Epidemiology

MASLD is becoming the most common liver disease worldwide. The increase in prevalence and severity is closely linked with the rise in obesity and metabolic syndrome worldwide. Locally, the prevalence was found to be up to 40% (Goh GB, 2016).

The material for subsequent sections has been adapted from various source including but not limited to (Muthiah, 2021), (Mantovani A, 2021) and the American Association for the Study of Liver Disease (AASLD) as well as in consultation with gastroenterologists from NUHS.

## **SCREENING AND DIAGNOSIS**

### **Screening**

Most patients with MASLD are usually asymptomatic and are picked up incidentally when they present with raised liver enzymes or hepatic steatosis on imaging performed for other reasons such as:

- Health screening
- Following commencement of medications for which routine liver enzyme monitoring is recommended (e.g. statins, allopurinol)

If presenting with raised liver enzymes, patients with MASLD usually have:

- ALT of between 40 to 250 U/L (Consider alternate pathology if the ALT is higher than this)
- ALT higher than AST
- Some may also have raised ALP
- Raised GGT (if done as part of external screening)

Certain patients are at increased risk of developing advanced liver disease. Targeted screening can be considered for at risk populations to identify these patients with clinically significant fibrosis ( $\geq$  F2 fibrosis) earlier.

## Diagnosis

- Hepatic steatosis<sup>1</sup> on imaging (or biopsy) AND
- At least 1 out of the 5 following cardiometabolic criteria,
  - BMI  $\geq 23$  kg/m<sup>2</sup> (or  $> 25$  kg/m<sup>2</sup> for Caucasian) OR Waist circumference  $> 94$  cm (Males) /  $80$  cm (Females)
  - Type 2 diabetes mellitus or Pre-diabetes.
  - Blood pressure  $\geq 130/85$ mmHg OR on specific treatment for hypertension
  - Plasma triglycerides  $\geq 1.70$  mmol/L OR on lipid lowering treatment.
  - Plasma HDL  $\leq 1.0$  mmol/L (Males) /  $\leq 1.3$  mmol/L (Females) OR on lipid lowering treatment.
- Secondary causes of hepatic steatosis should be ruled out. These include the below:
  - Large amount of alcohol consumption
    - Males  $> 21$  standard drinks<sup>2</sup> weekly
    - Females  $> 14$  standard drinks weekly
  - Viral hepatitis (Hepatitis B and Hepatitis C)
    - Hepatitis C is an unsubsidized test in NUP, consider testing if high risk behaviors present.<sup>3</sup>
  - Drug induced liver injury (DILI) from medication, complementary and alternative medicine (CAM) and supplements.
    - Refer to [LiverTox](#) for more information for suspected DILI.

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<sup>1</sup> Hepatic steatosis might be reported as fatty liver on ultrasound HBS report currently.

<sup>2</sup> Singapore 1 standard drink = 10 gram ethanol equivalent to 330ml of 5% ABV beer, 100ml of wine at 15% ABV, 30ml of spirits at 40% ABV

<sup>3</sup> Family history of Hepatitis C, intravenous drug use or sharing of sharps, tattoos, high risk sexual activity

## Initial Assessment

- MASLD may be suspected when patients present with hepatic steatosis on imaging and/or the typical pattern of raised liver enzymes suggestive of MASLD and if they fulfill at least 1 of the above cardiometabolic criteria.
- Secondary causes of raised liver enzymes should be considered and ruled out.
- Recommended evaluation as below:

|                           |                                                                                                                  |                             |                                                                                                                             |
|---------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| History                   | Alcohol intake<br>Concomitant medications<br>Complementary and alternative medications (CAM), herbs, supplements |                             |                                                                                                                             |
| Physical Examination      | Blood pressure<br>BMI<br>Waist circumference<br>Assess for hepatomegaly and stigmata of chronic liver disease    |                             |                                                                                                                             |
| Investigations            |                                                                                                                  |                             |                                                                                                                             |
| Imaging                   |                                                                                                                  | Ultrasound of the liver     |                                                                                                                             |
| Laboratory investigations | Comorbidities                                                                                                    |                             | HbA1c, lipid profile                                                                                                        |
|                           | Complications of chronic liver disease                                                                           |                             | FBC for thrombocytopenia<br>Liver function test (minimally AST / ALT)                                                       |
|                           | Other causes of raised liver enzymes                                                                             | Viral hepatitis             | HBsAg, Anti-HBs antibodies<br>Anti-HCV antibodies for high-risk patients                                                    |
|                           |                                                                                                                  | Wilson disease <sup>4</sup> | Strong family history of neurological or psychiatric history – refer to gastroenterology for evaluation of Wilson’s disease |

<sup>4</sup> Other possible causes of raised liver enzymes that are uncommon in the local population include autoimmune causes, hemochromatosis, celiac disease and alpha-1 anti-trypsin 1 deficiency. Refer **judiciously** to gastroenterology for workup.

## **Risk Stratification**

See management.

## **MANAGEMENT AND FOLLOW-UP**

### **Management In Primary Care**

- Lifestyle modifications
  - Abstain from regular alcohol.
    - Occasional 1-2 standard drinks per week is permissible for special occasions.
  - Engage in regular physical exercise.
- Management of metabolic comorbidities
  - Diabetes mellitus:
    - Screening for diabetes in patients with a new diagnosis of MASLD who have no known history of diabetes mellitus.
    - Management as per current NUP DM CPG / local guidelines.
  - Hypertension: Management as per current NUP CPG / Local guidelines.
  - Hyperlipidemia: Management as per current NUP CPG / Local guidelines.
  - Promote sustained weight loss for all patients who are overweight.
    - Encourage > 10% weight loss. Liver and cardiometabolic benefits begin at 5-7% of weight loss.
    - Achieve normal BMI of between 18.5 to 23 kg/m<sup>2</sup> in Asians (18.5 to 24.9 kg/m<sup>2</sup> in Caucasians).
- Vaccinations
  - Hepatitis A and B if non-immune.
  - Influenza vaccination yearly.
  - Pneumococcal vaccination.
    - Adults aged 18 to 64 years old: One dose of PPSV23.
    - Adults aged ≥ 65 years old: One dose of PCV13, one dose of PPSV23 1 year later.
- Cancer Screening
  - Given the increased risk of colon and breast cancer in MASLD patients, it is important to adhere to the current recommendations for cancer screening. There is insufficient evidence for hepatocellular carcinoma screening in these patients if they do not have cirrhosis.

- Fibrosis Risk Stratification

- Liver fibrosis is the key determinant of development of liver related complications and mortality.
- The gold standard of liver fibrosis is a biopsy. However, they are invasive, time consuming and has relatively higher risk of procedure related complications.
- The FIB-4 score is a non-invasive test developed originally to help predict fibrosis in patient with HIV and hepatitis C co-infection (Sterling RK, 2006 Jun).
  - FIB-4 became the non- invasive test of choice for MASLD because of the following.
    - Requires only simple blood test of platelets, AST and ALT that are available easily.
    - Large cross section studies established the ability of FIB-4 score to predict advanced fibrosis in MASLD patient with a high level of concordance (Siddiqui).

Formula<sup>5</sup> for calculating is 
$$\text{FIB-4} = \frac{\text{Age (years)} \times \text{AST (U/L)}}{\text{Platelet count (10}^9\text{)} \times \sqrt{\text{ALT (U/L)}}}$$

- Fib-4 risk results stratification and management.

| FIB-4 score     | Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FIB-4 <1.3      | Optimize metabolic and lifestyle in primary care. Repeat FIB-4 score. <ul style="list-style-type: none"> <li>• Annually if T2DM or ≥ 2 cardiometabolic criteria</li> <li>• Every 2 years if not T2DM and ≤ 2 cardiometabolic criteria</li> </ul>                                                                                                                                                                                                                            |
| FIB-4 1.3 -2.67 | Refer to gastroenterologist or for Vibration Controlled Transient Elastography (VCTE) <sup>6</sup> if available. <ul style="list-style-type: none"> <li>• If stiffness 8-10kPa, commence intensive lifestyle and diet changes to lose weight.</li> <li>• If stiffness &gt; 10kPa, refer to gastroenterologist for specialized management.               <ul style="list-style-type: none"> <li>○ Currently open access VCTE is not available at NUP.</li> </ul> </li> </ul> |
| FIB-4 > 2.6     | Refer to gastroenterologist                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

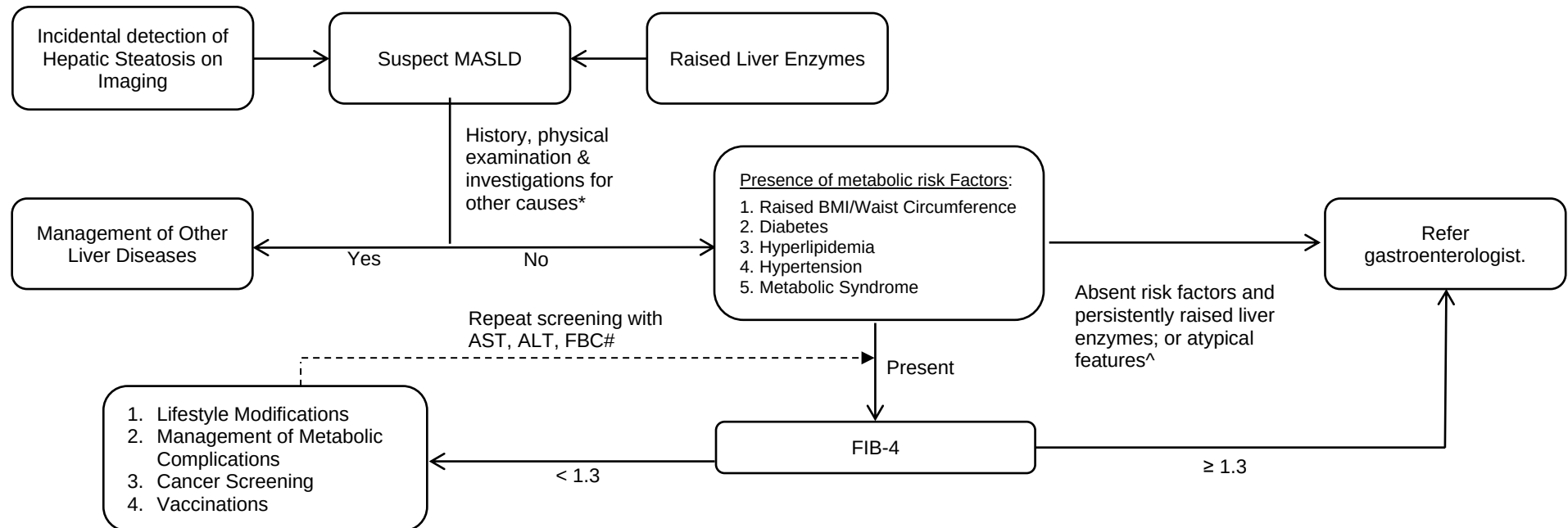
*Some practice points:*

*As the score is reliant on lab values, other conditions that affect AST, ALT and platelets count can make FIB-4 inaccurate. E.g. patient with low platelets from medication, recent infection or autoimmune thrombocytopenia can have a falsely elevated high FIB-4 score. Elevated AST/ALT from medication (e.g. statin initiation), alcohol or TCM consumption can also falsely elevated the FIB-4 score. The study was also conducted in patient around 40 years of age and has a higher rate of false positive in patient above the age of 65 (McPherson S, 2017 May).*

<sup>5</sup> Can be calculated online via [Fibrosis-4 \(FIB-4\) Index for Liver Fibrosis \(mdcalc.com\)](https://mdcalc.com). Epic enhancement for auto calculation and updates to [NUP Medical Calculator](#) to be announced.

<sup>6</sup> Fibroscan is a common type of VCTE currently in use within NUHS cluster.

## Management Algorithm



\*Other causes include alcohol, drugs, TCM, viral hepatitis, *autoimmune liver diseases*, *Wilson's disease*, and *Hemochromatosis*

#Repeat FIB4 every 1 year if T2DM or  $\geq 2$  metabolic risk factors; Repeat every 2 years if no T2DM and  $< 2$  metabolic risk factors

^Atypical features:  $\geq$  family members with idiopathic / cryptogenic cirrhosis, features suggestive of Wilson's disease e.g. family history of neurological or psychiatric illness

### **Treatment Goals / Targets of Control**

See above management.

### **Re-Assessment and Annual Assessment**

See algorithm.

### **Non-Pharmacological / Lifestyle**

See management.

### **Role of Non-Doctor Team Members**

1. Dietician should be involved for dietary counselling to aid in management of metabolic conditions, especially with regards to weight loss.
2. Care managers should provide adjuvant counselling for management of metabolic conditions.
3. Care coordinators should encourage uptake of preventive health measure such as vaccinations and cancer screening.

### **REFERRALS AND COMMUNITY RESOURCES**

#### **Referrals to Hospital**

See management and algorithm.

### **RECOMMENDED CARE COMPONENTS**

1. Annual FIB-4 scoring.
2. Good control of metabolic conditions – namely Hypertension, Diabetes, Pre- Diabetes, BMI and Hyperlipidaemia.
3. Up to date and appropriate vaccinations.

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1. Goh GB, K. C. (2016). Perceptions of non-alcoholic fatty liver disease - an Asian community-based study. *Gastroenterology report*, 2016;4(2):131-5.
2. Mantovani A, P. G. (2021). Non-alcoholic fatty liver disease and risk of incident diabetes mellitus: an updated meta-analysis of 501 022 adult individuals. *Gut*, 70(5):962-969.
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5. Siddiqui, M. S. (n.d.). Diagnostic Accuracy of Noninvasive Fibrosis Models to Detect Change in Fibrosis Stage. *Clinical Gastroenterology and Hepatology*, Volume 17, Issue 9, 1877 - 1885.e5.
6. Sterling RK, L. E. (2006 Jun). Clinical Investigators. Development of a simple noninvasive index to predict significant fibrosis in patient. *Hepatology*. 2006 Jun;43(6), 43(6):1317-25.

## **ANNEX 1: MANAGEMENT OF ELEVATED ALANINE TRANSAMINASE (ALT)**

**Table 1: Referral Criteria for Raised ALT**

|                                                                                                                                                                                                                                                                          |                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1. The patient has elevated ALT of > 1000 IU.<br>2. The patient has elevated ALT of any level and has any one of the following:<br>a) Signs of acute liver failure<br>b) Jaundice with fever<br>c) Clinically ill                                                        | Refer A&E              |
| If the patient has elevated ALT, is <b>clinically well</b> and has any one of the following:-                                                                                                                                                                            |                        |
| 1. $200 \leq \text{ALT} \leq 1000$ IU<br>2. Solid mass (liver, pancreas, intra-abdominal) on imaging<br>3. Jaundice with no fever<br>4. Hepatomegaly, splenomegaly, ascites, oedema                                                                                      | Refer GE Direct Access |
| 5. Cirrhosis - suspected or newly diagnosed as indicated by<br>a) raised bilirubin without clinical jaundice;<br>b) low albumin<br>c) low platelets<br>6. Transaminitis (Non-Hep B) with elevated globulin<br>7. Unexplained weight loss ( $\geq 5\%$ in 6 to 12 months) | Refer GE Early         |
| 8. Transaminitis (Non-Hep B) i.e., persistently elevated ALT ( $120 \leq \text{ALT} < 200$ IU) for at least 2 weeks<br>9. Hepatitis C                                                                                                                                    | Refer GE Routine       |

|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STEP 1</b><br><br>Check the ALT level and review patient condition, signs and symptoms and any other available investigations results or history. Refer to Table 1 for referral criteria.                                                                                                                                                                                                                          | ALT > 1000                                                                                  |                                                                                                                                                                                                                                                                                                                                                            | Refer A&E                                                                                                                                                                                                                                                                                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | 200 ≤ ALT ≤ 1000 IU                                                                         |                                                                                                                                                                                                                                                                                                                                                            | Refer GE Direct Access K.I.V. A&E if patient is clinically ill                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | Investigate for causes if ALT is elevated but below 200 IU.<br>i.e. (Normal < ALT < 200 IU) |                                                                                                                                                                                                                                                                                                                                                            | 1. Review drug and alcohol use – Withdraw offending agent, manage alcohol dependence and repeat LFT in 4 weeks                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            | 2. Review Hep B Status, if Positive – Manage according to Hepatitis B guideline                                                                                                                                                                                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            | 3. Review Hep C status, if Positive – Refer GE                                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            | 4. If <b>BOTH</b> status for Hep B and Hep C are unknown, offer:<br>a) Hep B screening, and manage accordingly<br>b) Hep C screening (Test is <b>not</b> subsidised), and manage accordingly<br>c) K.I.V. to include LFT when offering above screening i.e., Proceed to <b>STEP 2</b>                                    |
| <b>STEP 2</b><br><br>Repeat LFT if patient is clinically well and is KNOWN to be negative for both Hep B and C; and there is no known cause for the elevated ALT: -<br><br>1. Repeat LFT in 1-2 weeks if ALT is <b>120 ≤ ALT ≤ 200 IU</b><br><b>OR</b><br>2. Repeat LFT in 1-2 months if ALT is <b>Normal &lt; ALT &lt; 120 IU</b><br><br>Manage based on repeat ALT level. Refer to <b>A, B or C</b> in next column. | <b>A</b>                                                                                    | If repeat ALT is ≥ 120 IU                                                                                                                                                                                                                                                                                                                                  | Refer to <a href="#">Table 1</a>                                                                                                                                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>B</b>                                                                                    | If repeat ALT is Normal                                                                                                                                                                                                                                                                                                                                    | Repeat LFT in 3-4 months: -<br>a) if ALT remains normal – <b>Discharge</b><br>b) If ALT ≥ 120 IU – Refer to <a href="#">Table 1</a><br>c) if ALT elevated but < 120 IU – Proceed to <b>C</b> below                                                                                                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>C</b>                                                                                    | If repeat ALT is raised but < <b>120 IU</b> , to proceed with the following tests.<br>a) FBC<br>b) Thyroid Function Test<br>c) Baseline U/S liver<br><br><b>* If all the FBC, TFT and U/S Liver are normal, repeat LFT 1-2 months and manage: -</b><br>a) If ALT is ≥ 120 IU - Refer to <b>Table 1</b><br>b) If ALT is < 120 IU - <b>Proceed to STEP 3</b> | 1. Refer to <a href="#">Table 1</a> If any <b>ONE</b> of following is present:<br>a) Thrombocytopenia<br>b) U/S – liver cirrhosis / coarse or nodular echotexture / HCC                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            | 2. If thyroid dysfunction present – Treat and <b>THEN</b> repeat LFT in 3-4 months: -<br>a) If ALT is ≥ 120 IU Refer to <a href="#">Table 1</a><br>b) If ALT is < 120 IU, repeat LFT in 1-2 months, and: -<br>i. If ALT ≥ 120 IU – Refer to <a href="#">Table 1</a><br>ii. If ALT is < 120 IU – <b>Proceed to STEP 3</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            | 3. If U/S Liver is Fatty liver or showed increased echogenicity with no other liver abnormalities, treat as for Fatty Liver                                                                                                                                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                             |                                                                                                                                                                                                                                                                                                                                                            | 4. If U/S is Normal – <b>Proceed to STEP 3</b>                                                                                                                                                                                                                                                                           |
| <b>STEP 3</b><br>Repeat LFT in 3-4 months                                                                                                                                                                                                                                                                                                                                                                             | a. if ALT is ≥ 120 IU                                                                       |                                                                                                                                                                                                                                                                                                                                                            | Refer to <a href="#">Table 1</a>                                                                                                                                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | b. if ALT is < 120 IU and patient is well                                                   |                                                                                                                                                                                                                                                                                                                                                            | K.I.V. repeat in 6 to 12 months or <b>Discharge</b>                                                                                                                                                                                                                                                                      |